

DevOps Engineer — 90 Minute Infrastructure Challenge

Time Limit: 90 minutes total

Video Length: 8–12 minutes

Task

Build and deploy a minimal production-style application stack using Kubernetes and CI/CD.

Your system should demonstrate:

- containerization,
- deployment automation,
- observability,
- and operational debugging ability.

You may use any tools/frameworks/cloud providers locally or remotely.

Requirements

1. Working Deployment (Mandatory)

Deploy a simple application stack to Kubernetes.

Minimum requirements:

- 1 backend service
- 1 database/service dependency
- Kubernetes deployment working end-to-end

You may use:

- Minikube
- Kind
- k3s
- EKS/AKS/GKE
- or any equivalent setup

The application itself can be extremely simple.

We are evaluating infrastructure quality, not frontend complexity.

2. CI/CD Pipeline (Mandatory)

Create a working CI/CD pipeline that:

- builds the application,
- creates a Docker image,
- and deploys to Kubernetes automatically.

You may use:

- GitHub Actions
 - GitLab CI
 - Jenkins
 - ArgoCD
 - or equivalent tooling.
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3. One Real Reliability Improvement (Mandatory)

Choose ONE and implement it properly:

- readiness/liveness probes
- autoscaling
- rollback handling
- Helm templating
- secret management
- logging/monitoring
- resource tuning
- canary/blue-green deployment
- persistent storage handling
- ingress/routing
- retry/circuit breaker logic

You must explain:

- why you chose it,
- what problem it solves,

- and what tradeoff it introduces.
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4. Intentional Failure Simulation (Mandatory)

Intentionally create ONE failure scenario and debug it live.

Examples:

- crashing container
- failed health check
- bad environment variable
- image pull issue
- broken ingress
- CPU/memory issue
- failed deployment
- database connectivity issue
- incorrect Kubernetes config

You must:

- show the failure,
- debug it,
- explain your reasoning,
- and fix it.

This is the most important section.

Submission Requirements

Video Submission (Mandatory)

1. Live Demo (3–4 min)

Show:

- application working,
- Kubernetes resources,
- CI/CD execution,

- and successful deployment.
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2. Architecture Walkthrough (2–3 min)

Explain:

- cluster setup,
 - deployment flow,
 - CI/CD pipeline,
 - reliability decisions,
 - and why you structured things this way.
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3. Failure Debugging Walkthrough (2–3 min)

Show:

- the exact failure,
- symptoms observed,
- commands/logs/metrics checked,
- wrong assumptions (if any),
- and final root cause.

We care heavily about debugging methodology.

4. Tradeoff Discussion (1–2 min)

Explain:

- what you intentionally simplified,
 - what would break at scale,
 - and what you would improve in a real production environment.
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Rules

- Do NOT use fully managed “one-click” deployment platforms
- Do NOT hide Kubernetes details behind abstractions